

Practical 4: Trend Analysis and Transformation of US Population Dataset

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1. Question

For a given dataset of the US Population (in millions), write the R code to:

- (a) Import the data as a time series object.
- (b) Identify the dominating component(s) in the data set.
- (c) Apply a square root transformation to the original data set.
- (d) Estimate the linear trend present in the data set.
- (e) Remove the estimated linear trend from the data set.

2. Objective

To analyze US Population data by converting it into a time-series object, applying a variance-stabilizing square root transformation, and performing detrending to isolate the residual fluctuations from the linear trend.

3. R Code

```
# Clear workspace
rm(list = ls(all = TRUE))

# Load libraries
library(tseries)
library(utils)
library(readr)

# (a) Create time series object
year <- 1970:1990

population <- c(
  3929214, 5308483, 7239881, 9638453, 12860702,
  17063353, 23191876, 31443321, 38558371, 50189209,
  62979666, 76212168, 92228496, 106021537, 123202624,
```

```

132164569, 151325798, 179323175, 203302031, 226542203,
248709873
)

pop_ts <- ts(population, start = 1970, end = 1990, frequency = 1)

plot(
  pop_ts,
  main = "US Population Time Series",
  ylab = "Population",
  xlab = "Year",
  col = "blue",
  lwd = 2
)

```

```

# (b) Dominating component: Upward Trend

```

```

# (c) Square root transformation

```

```

pop_sqrt <- sqrt(pop_ts)

plot(
  pop_sqrt,
  main = "Square Root Transformed Population",
  ylab = "Sqrt(Population)",
  xlab = "Year",
  col = "red",
  lwd = 2
)

```

```

# (d) Estimate linear trend using regression

```

```

time <- time(pop_ts)
trend_model <- lm(pop_ts ~ time)
trend_estimated <- trend_model$fitted.values

plot(
  pop_ts,
  main = "Population with Linear Trend",
  ylab = "Population",
  xlab = "Year",
  col = "blue",
  lwd = 2
)
lines(trend_estimated, col = "darkgreen", lwd = 2)

```

```

# (e) Remove estimated linear trend

```

```

pop_detrended <- pop_ts - trend_estimated

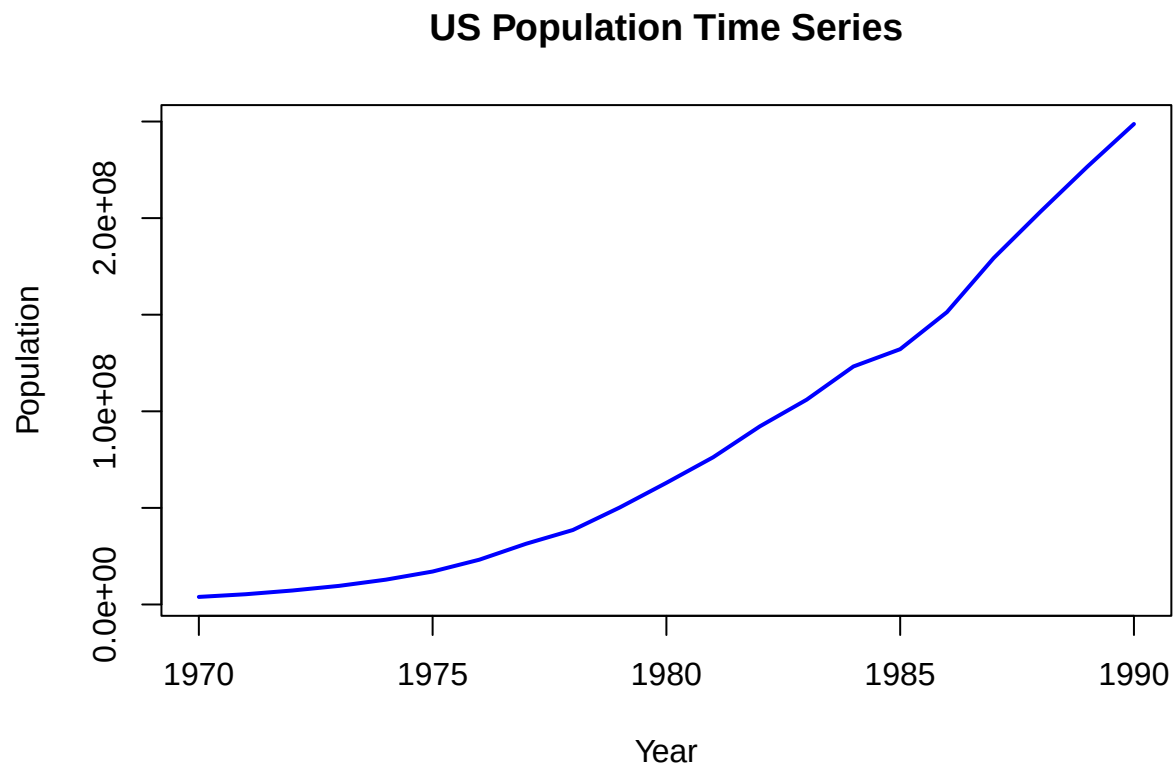
plot(
  pop_detrended,
  type = "o",
  main = "Detrended Population Series",
  ylab = "Detrended Values",
  xlab = "Year",

```

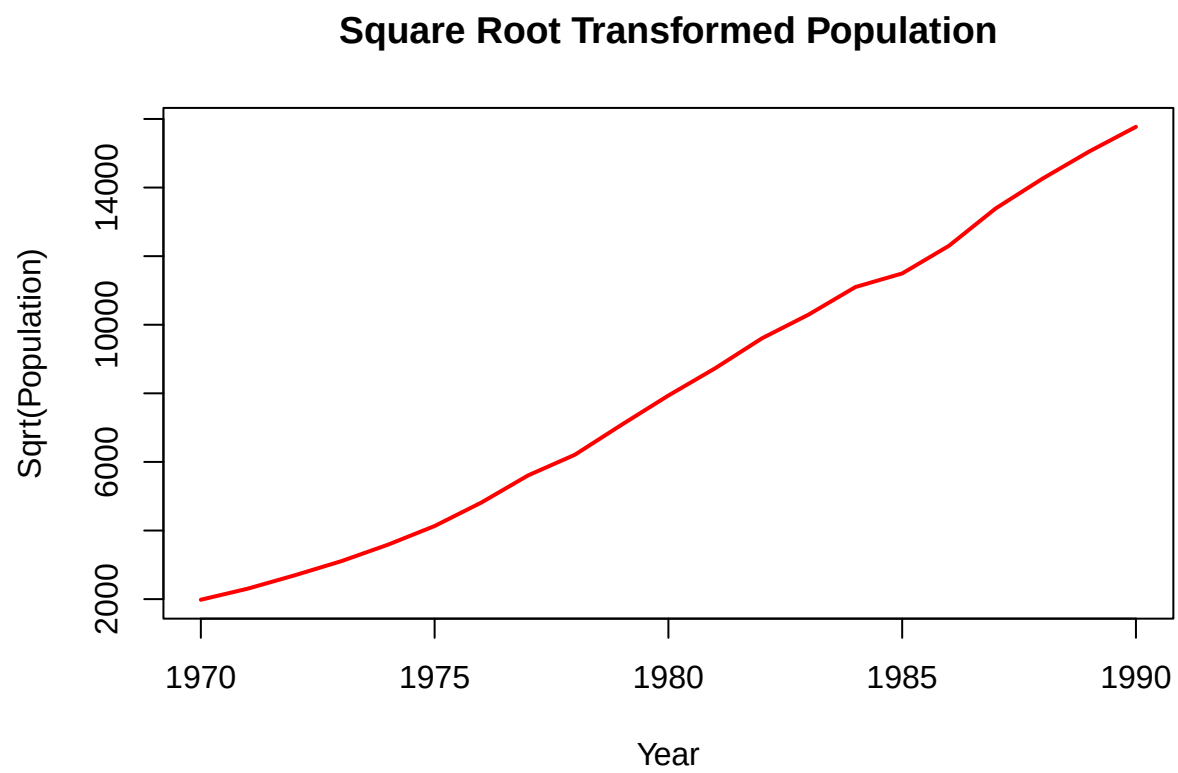
```
col = "purple",  
lwd = 2  
)
```

4. Output

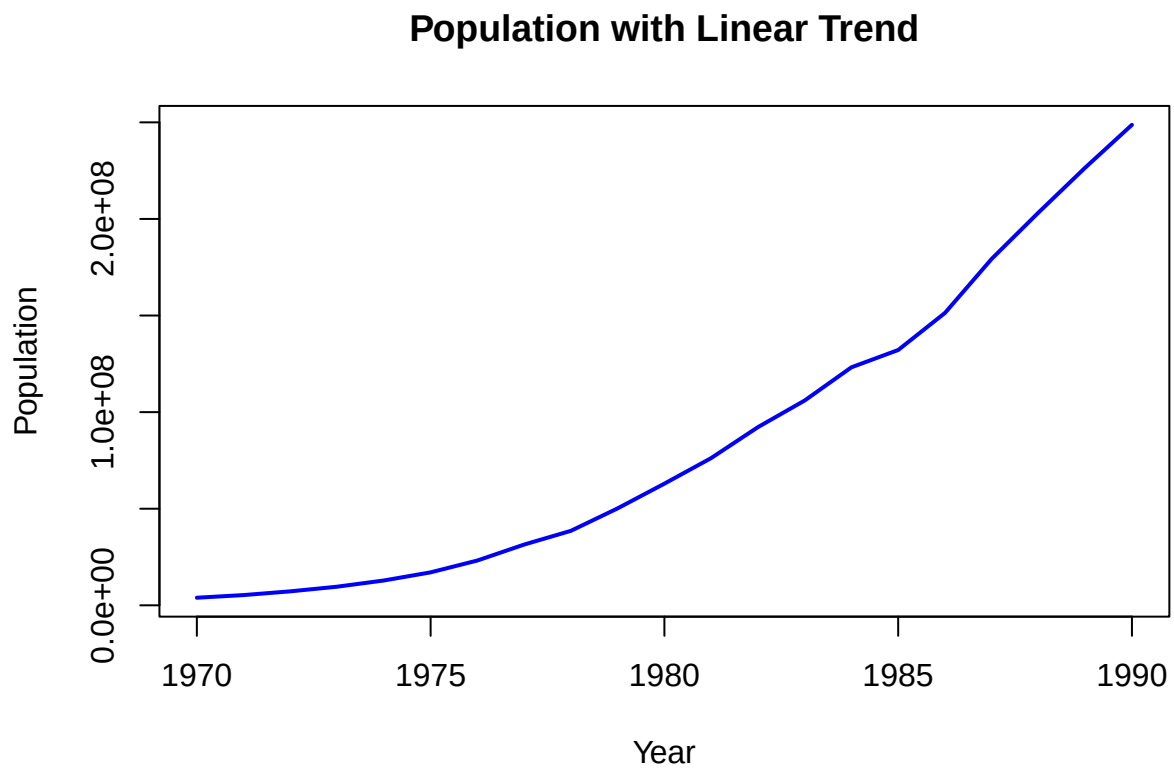
Original Time Series



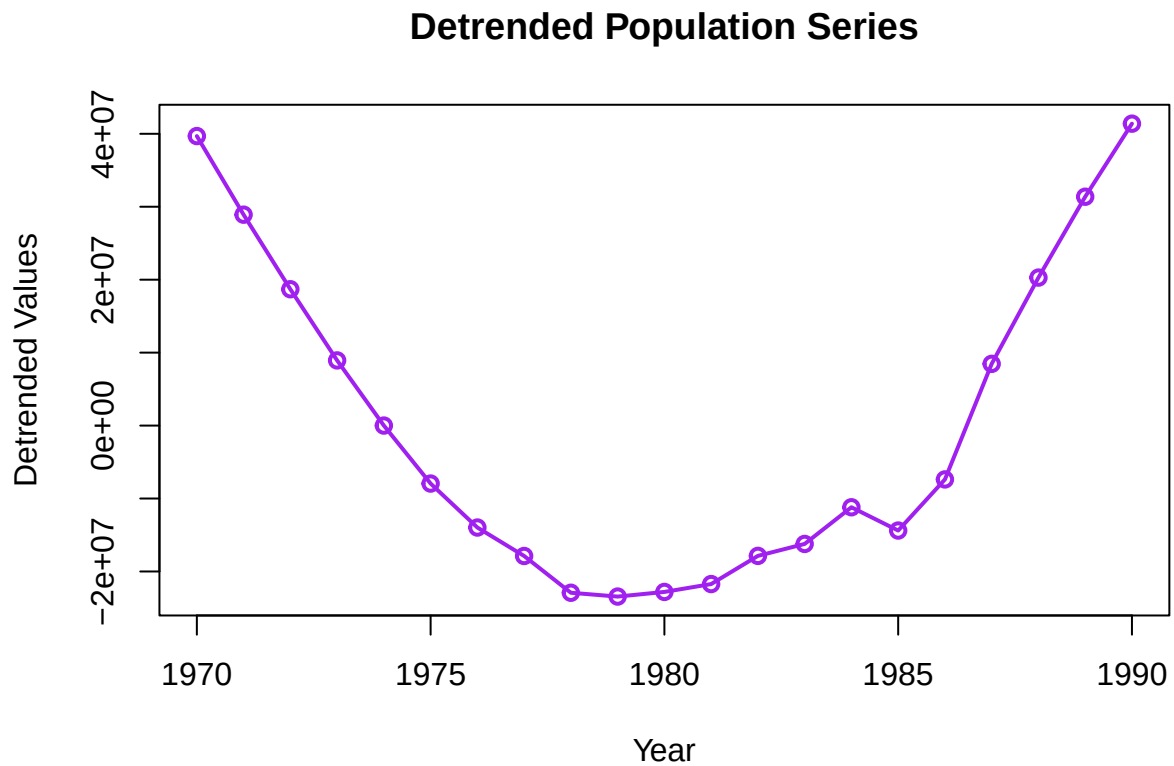
Square Root Transformed Series



Population with Linear Trend



Detrended Series



5. Conclusion

In this practical, we analyzed the US Population dataset and identified that the Trend component is dominant, as the data shows a consistent upward movement over time.

We successfully applied a square root transformation to stabilize the variance of the series. A linear regression model was then used to estimate the linear trend component.

By subtracting the estimated trend from the original time series, we obtained the detrended series. This detrended data allows us to analyze fluctuations around the long-term growth pattern and better understand the underlying stationary behavior of the dataset.